Muhammad Ashir

DATA SCIENTIST / AI SYSTEMS ENGINEER

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PROFILE

Data Science master's student and software engineer with 2+ years of experience in banking and FinTech. Experienced in Oracle-based enterprise systems, ETL pipelines, data modeling, and KYC workflows, with a 40% database performance improvement delivered in a core-banking environment. Current work focuses on responsible AI, financial LLM systems, explainable risk, and secure agentic payments.

PROFESSIONAL EXPERIENCE

Software Engineer
JUL 2024 — SEP 2025

Bootlogix — Pakistan

- Optimized ETL pipelines and core-banking data extraction, improving processing reliability and accuracy.
- Improved system performance by 40% through database query optimization and data modeling while maintaining data integrity.
- Managed Oracle applications and servers, supporting high system uptime and reduced operational interruptions.
- Collaborated across functions to deliver solutions 15% faster and led legacy-system migration to modern infrastructure.

Software Engineer
SEP 2023 — MAY 2024

CubeBytes Solutions — Pakistan

- Contributed to a Central Bank of Egypt project by optimizing Oracle-based enterprise solutions for scalable financial systems.
- Enhanced ETL processes across critical banking modules with the DBA team.
- Installed, configured, and maintained Oracle Analytics Server for reliable reporting and analytics.
- Redesigned KYC algorithms and workflows to strengthen regulatory compliance.

SELECTED PROJECTS

MandatePay — Authorization Firewall for Agentic Payments
2026

Python · MCP · Policy-as-code · HMAC · Hash-chained audit

- Built a research-grade control plane that verifies machine-initiated payments against signed human mandates.
- Implemented deterministic policy limits, agent identity binding, human approval challenges, replay safety, and verifiable decisions.

RegLedger AI — Evidence-first Compliance Intelligence
2026

Python · RAG · Explainable risk · Audit chain

- Built transparent transaction triage with named risk signals, evidence, recommended controls, and tamper-evident audit events.
- Implemented policy-grounded retrieval with an optional LLM provider and deterministic offline fallback.

Europulse Fraud Guard
2026

PyTorch · Graph Attention Networks · FastAPI

- Designed a dual-stream fraud-detection architecture combining graph intelligence, behavioral signals, and explainable reason codes.

EDUCATION

Master of Data Science
2025 — PRESENT

Technische Universität Ilmenau — Germany

- Advanced studies in machine learning, statistical modeling, big data analytics, and data-driven system development.

Bachelor of Science in Software Engineering
2020 — 2024

University of Central Punjab — Pakistan

- Object-oriented programming, data structures and algorithms in C++, software design, and design patterns.

TECHNICAL SKILLS

Programming: Python, SQL, Java, JavaScript, C++, HTML, CSS
Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, XAI, RAG, EDA, data modeling
Data & Infrastructure: Oracle, ETL pipelines, FastAPI, Streamlit, Jupyter, Git, Linux, VS Code
Languages: English (fluent), German (beginner)